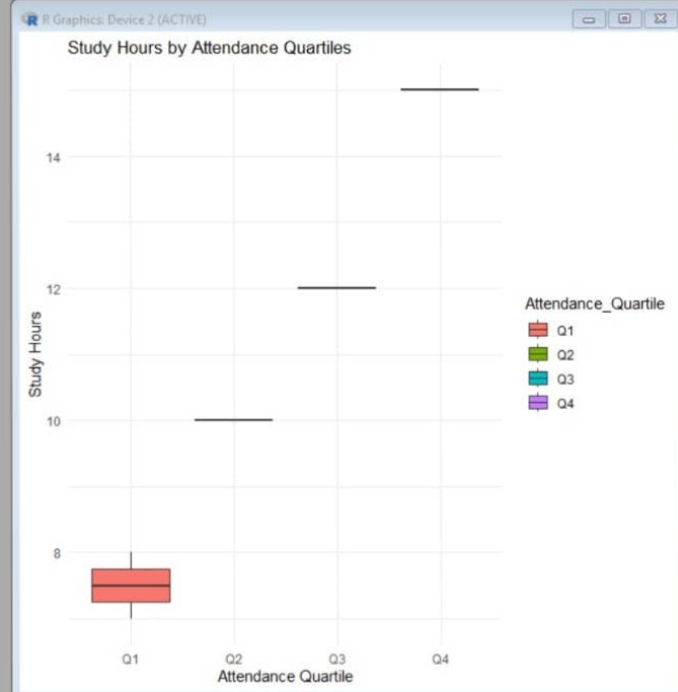


```
> Age = c(19,21,20,22,23),
+ Study_Hours = c(12,8,15,10,7),
+ Attendance = c(90,70,95,85,60),
+ Test_Score = c(85,70,92,80,65),
+ Participation_Score = c(8,7,9,8,6)
+ )
> area_data <- pivot_longer(
+   student,
+   cols = c(Test_Score, Participation_Score),
+   names_to = "variable",
+   values_to = "value"
+ )
> ggplot(area_data, aes(x = Student_ID, y = value, fill = variable)) +
+   geom_area(stat = "identity", position = "stack") +
+   labs(title = "Test Score & Participation Score Trend",
+        x = "Student",
+        y = "Score") +
+   theme_minimal()
> ggplot(area_data, aes(x = Student_ID, y = value, fill = variable)) +
+   geom_bar(stat = "identity", position = "stack") +
+   labs(title = "Test Score & Participation Score Trend",
+        x = "Student",
+        y = "Score") +
+   theme_minimal()
>
4
```



```
> )
> ggplot(area_data, aes(x = Student_ID, y = value, fill = variable)) +
+   geom_area(stat = "identity", position = "stack") +
+   labs(title = "Test Score & Participation Score Trend",
+         x = "Student",
+         y = "Score") +
+   theme_minimal()
> ggplot(area_data, aes(x = Student_ID, y = value, fill = variable)) +
+   geom_bar(stat = "identity", position = "stack") +
+   labs(title = "Test Score & Participation Score Trend",
+         x = "Student",
+         y = "Score") +
+   theme_minimal()
> student$Attendance_Quartile <- cut(student$Attendance,
+                                   breaks = quantile(student$Attendance),
+                                   include.lowest = TRUE,
+                                   labels = c("Q1", "Q2", "Q3", "Q4"))
>
> ggplot(student, aes(x = Attendance_Quartile, y = Study_Hours, fill = Attendance_Quartile)) +
+   geom_boxplot() +
+   labs(title = "Study Hours by Attendance Quartiles",
+         x = "Attendance Quartile",
+         y = "Study Hours") +
+   theme_minimal()
>
>
>
```



```
R Console
> 
> ggplot(df_long, aes(x = Student_ID, y = value, fill = variable, group = variable)) +
+   geom_area(alpha = 0.6, position = "stack") +
+   labs(title = "Stacked Area Chart: Test Score vs Participation Score",
+         x = "Students",
+         y = "Scores") +
+   theme_minimal()
Error: object 'df_long' not found
> student$Attendance_Quartile <- cut(student$Attendance,
+                                   breaks = quantile(student$Attendance),
+                                   include.lowest = TRUE,
+                                   labels = c("Q1", "Q2", "Q3", "Q4"))
> ggplot(student, aes(x = Attendance_Quartile, y = Study_Hours, fill = Attendance_Quartile)) +
+   geom_boxplot() +
+   labs(title = "Study Hours by Attendance Quartiles",
+         x = "Attendance Quartiles",
+         y = "Study Hours") +
+   theme_minimal()
> ggplot(student, aes(x = Test_Score)) +
+   geom_density(fill = "skyblue", alpha = 0.5) +
+   labs(title = "Density Plot of Test Scores",
+         x = "Test Score",
+         y = "Density") +
+   theme_minimal()
> 
```

